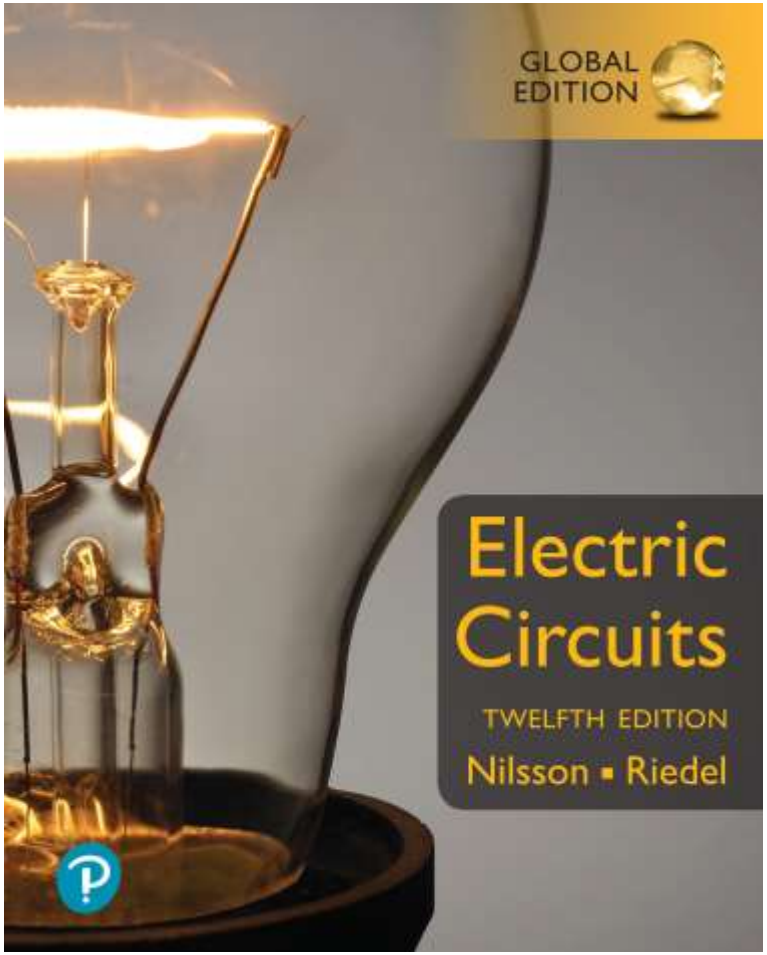
Electric Circuits

Twelfth Edition, Global Edition



Chapter 4 – Part 3

Techniques of Circuit Analysis

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Chapter 4 – techniques of circuit analysis (1 of 13)

- The mesh current method:
  - The **dual of the node voltage method**
  - Uses **KVL equations around meshes**
  - Solves **directly for currents**
  - **Special cases for dependent sources and for current sources in a mesh.**

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Chapter 4 – techniques of circuit analysis (2 of 13)

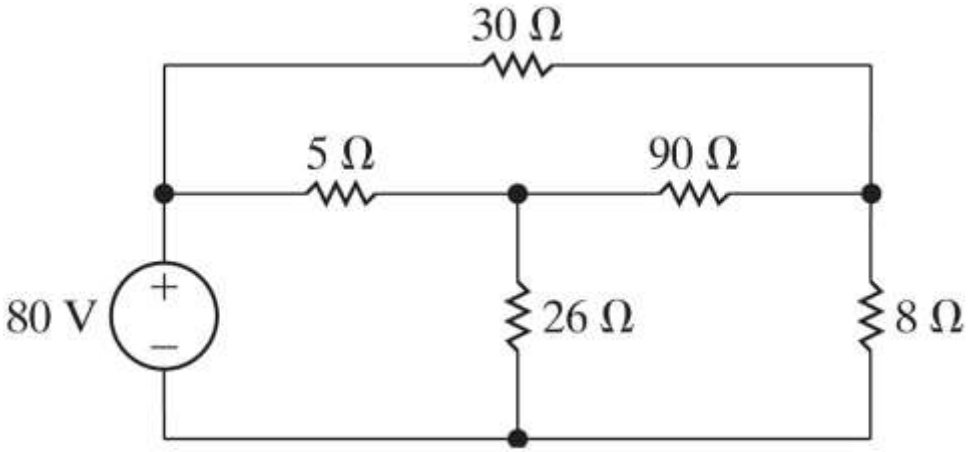
- The basic mesh current method recipe:
  - Step 1 – Identify the meshes with curved directed arrows that follow the perimeter of each mesh.
  - Step 2 – Label the mesh currents for each mesh.
  - Step 3 – Write a KVL equation for each mesh.
  - Step 4 – Solve the KVL equations to find the mesh current values.
  - Step 5 – Solve the circuit using mesh currents from Step 4 to find component currents, voltages, and power values.

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Chapter 4 – techniques of circuit analysis (3 of 13)

How many meshes does this circuit contain?



A. 1

B. 2

C. 3

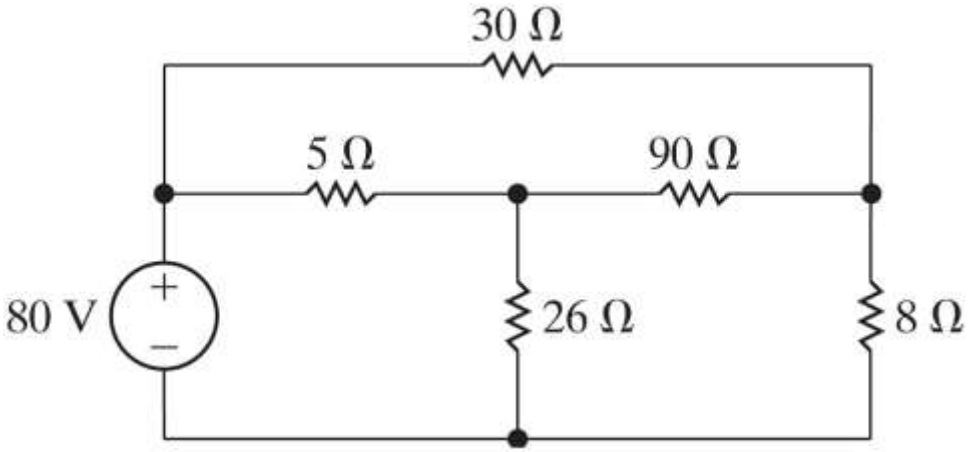
D. 4

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Chapter 4 – techniques of circuit analysis (3 of 13)

How many meshes does this circuit contain?



✗ A. 1

✗ B. 2

✓ C. 3

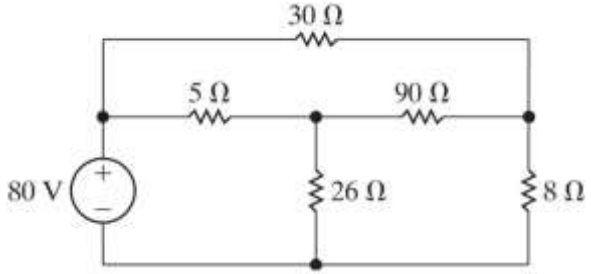
✗ D. 4

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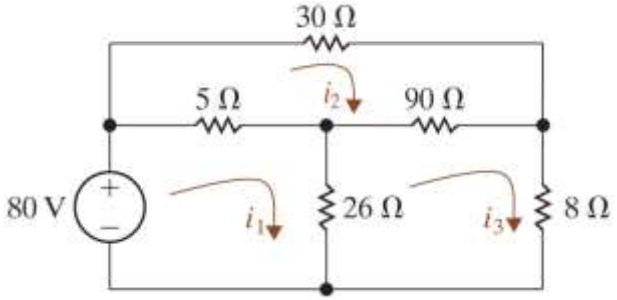
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Chapter 4 – techniques of circuit analysis (4 of 13)

Find the power associated with the voltage source and the 8Ω resistor, using the mesh current method.



After Step 1 (identifying the meshes) and Step 2 (labeling the mesh currents):



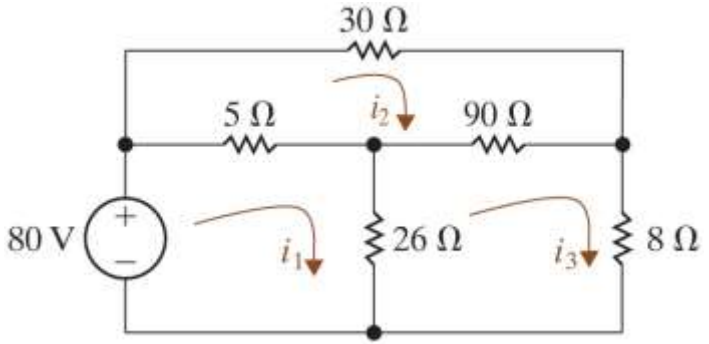
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Chapter 4 – techniques of circuit analysis (5 of 13)

Find the power associated with the voltage source and the $8\,\Omega$ resistor, using the mesh current method.

- Step 3 – Write a KVL equation for each mesh.
- Step 4 – Solve the KVL equations to find the mesh current values.



i_1 mesh: $-80 + 5(i_1 - i_2) + 26(i_1 - i_3) = 0$

i_2 mesh: $30(i_2) + 90(i_2 - i_3) + 5(i_2 - i_1) = 0$

i_3 mesh: $8(i_3) + 26(i_3 - i_1) + 90(i_3 - i_2) = 0$

Standard form:

$$\begin{aligned} i_1(5+26) + i_2(-5) + i_3(-26) &= 80 \\ i_1(-5) + i_2(30+90+5) + i_3(-90) &= 0 \\ i_1(-26) + i_2(-90) + i_3(8+26+90) &= 0 \end{aligned}$$

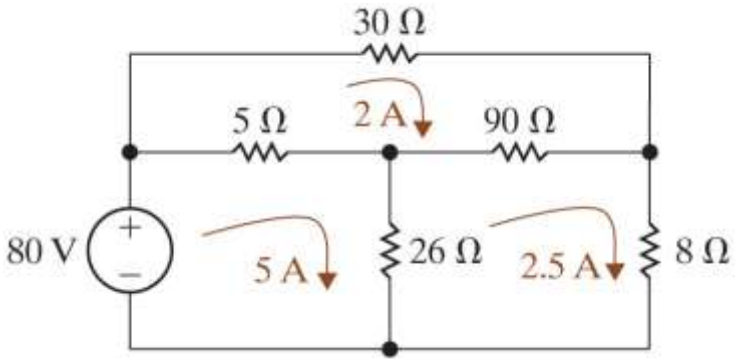
Solution: $i_1 = 5$ A; $i_2 = 2$ A; $i_3 = 2.5$ A

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Chapter 4 – techniques of circuit analysis (6 of 13)

- Step 5 – Solve the circuit using mesh currents from Step 4 to find component currents, voltages, and power values. Here we check that the power balances



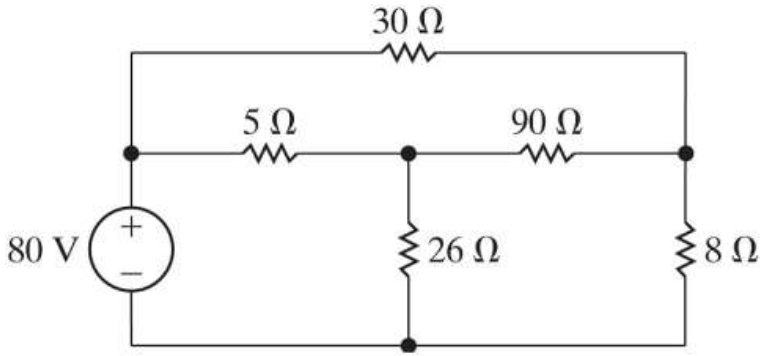
Component	Equation	p[W]
80 V	$-(5)(80)$	**−400**
5 Ω	$(5-2)^2(5)$	45
90 Ω	$(2.5-2)^2(90)$	22.5
30 Ω	$(2)^2(30)$	120
26 Ω	$(5-2.5)^2(26)$	162.5
8 Ω	$(2.5)^2(8)$	**50**

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Chapter 4 – techniques of circuit analysis (7 of 13)

If you were asked to use the Node Voltage method, how many equations would you write and solve?



A. 3 KCL, 0 constraint

B. 2 KCL, 1 constraint

C. 1 KCL, 2 constraint

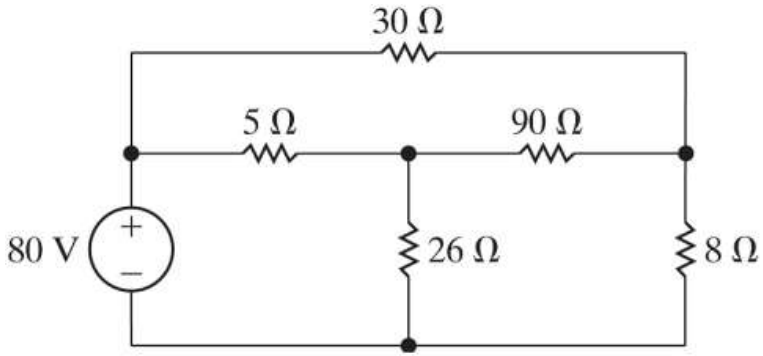
D. 2 KCL, 0 constraint

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Chapter 4 – techniques of circuit analysis (7 of 13)

If you were asked to use the Node Voltage method, how many equations would you write and solve?



- ✗ A. 3 KCL, 0 constraint
- ✗ B. 2 KCL, 1 constraint
- ✗ C. 1 KCL, 2 constraint
- ✓ D. 2 KCL, 0 constraint

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Chapter 4 – techniques of circuit analysis (8 of 13)

- Mesh current method, special cases:
  - Dependent sources
  - Current source on the perimeter of a mesh
  - Current source shared between two meshes

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Chapter 4 – techniques of circuit analysis (9 of 13)

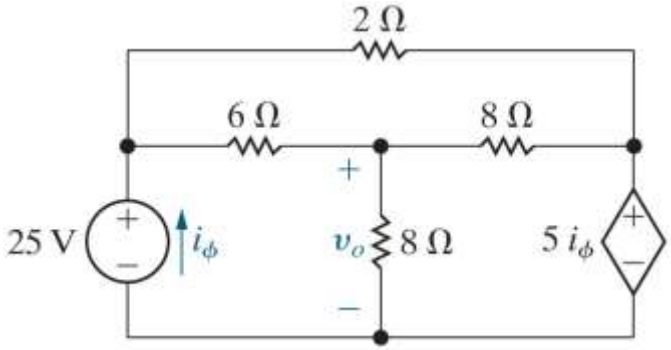
- Modify the mesh current analysis method to handle circuits with dependent sources:
  - Step 1 – Identify the meshes with curved directed arrows that follow the perimeter of each mesh.
  - Step 2 – Label the mesh currents for each mesh.
  - Step 3 – Write a KVL equation for each mesh.
    - If there are dependent sources, write a dependent source constraint equation for each.
  - Step 4 – Solve the KVL equations to find the mesh current values.
  - Step 5 – Solve the circuit using mesh currents from Step 4 to find component currents, voltages, and power values.

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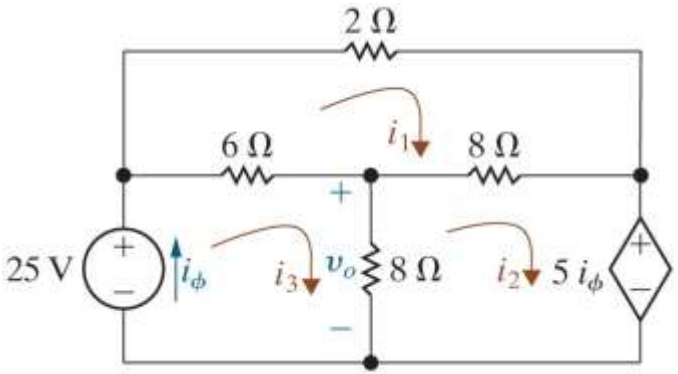
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Chapter 4 – techniques of circuit analysis (10 of 13)

Find v_o using the mesh current method.



After Steps 1 and 2:



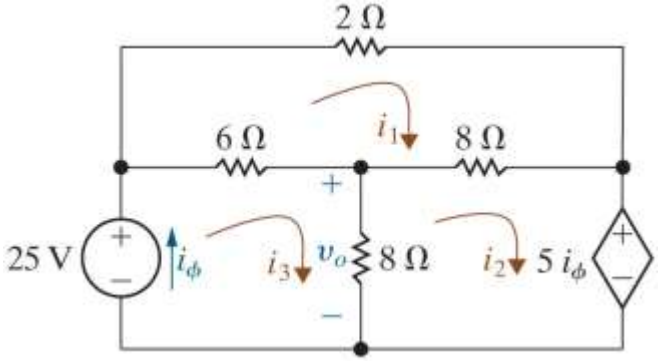
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Chapter 4 – techniques of circuit analysis (11 of 13)

Find v_o using the mesh current method.

Write the KVL and dependent source equations (Step 3) and solve them (Step 4).



i_1 mesh: $2i_1 + 8(i_1 - i_2) + 6(i_1 - i_3) = 0$

i_2 mesh: $5i_\phi + 8(i_2 - i_3) + 8(i_2 - i_1) = 0$

i_3 mesh: $-25 + 6(i_3 - i_1) + 8(i_3 - i_2) = 0$

constraint: $i_\phi = i_3$

Standard form:

$$\begin{aligned} i_1(2+8+6) + i_2(-8) + i_3(-6) &= 0 \\ i_1(-8) + i_2(8+8) + i_3(5-8) &= 0 \\ i_1(-6) + i_2(-8) + i_3(6+8) &= 25 \end{aligned}$$

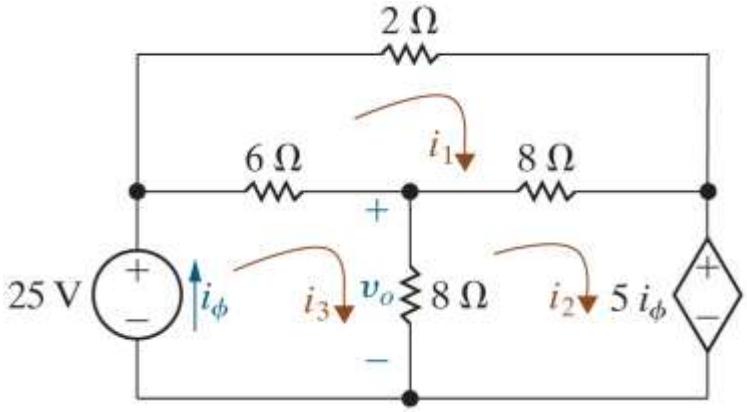
Solution: $i_1 = 2.5$ A; $i_2 = 2$ A; $i_3 = i_\phi = 4$ A

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Chapter 4 – techniques of circuit analysis (12 of 13)

If $i_1 = 2.5\,\text{A}, i_2 = 2\,\text{A}$, and $i_3 = 4\,\text{A}$ find v_o.



A. 16 V

B. 32 V

C. 2 V

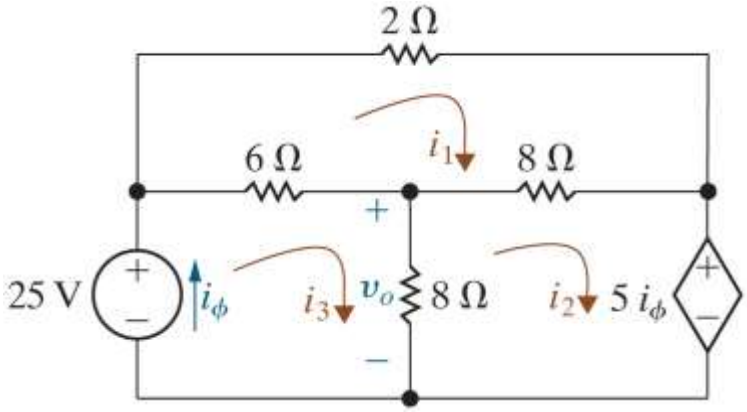
D. −16 V

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Chapter 4 – techniques of circuit analysis (12 of 13)

If $i_1 = 2.5\,\text{A}, i_2 = 2\,\text{A}$, and $i_3 = 4\,\text{A}$ find v_o.



- ✔ A. 16 V
- ✘ B. 32 V
- ✘ C. 2 V
- ✘ D. −16 V

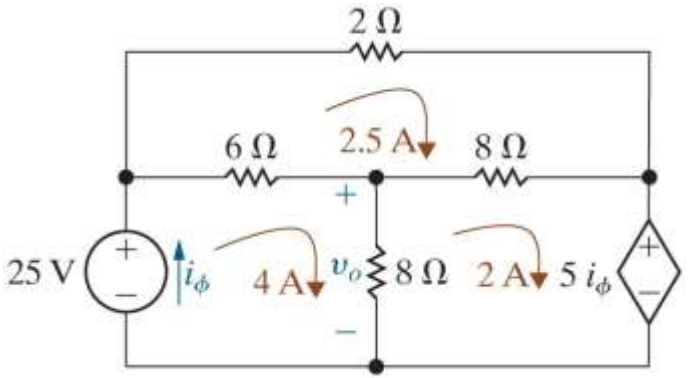
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Chapter 4 – techniques of circuit analysis (13 of 13)

Step 5 – Solve the circuit using mesh currents from Step 4 to find component currents, voltages, and power values.

$$v_o = 8(4-2) = 16 \text{ V}.$$



Power balance:

Component	Equation	p [W]
25 V	−(4)(25)	−100
Dep. Source	(2) [5(4)]	40
6 Ω	$(4-2.5)^2(6)$	13.5
2 Ω	$(2.5)^2(2)$	12.5
8 Ω (middle)	$(4-2)^2(8)$	32
8 Ω (right)	$(2-2.5)^2(8)$	2

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